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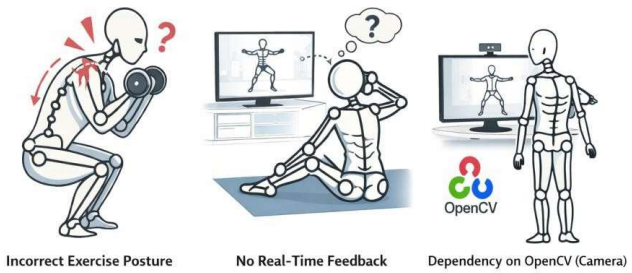
Title of Project : Fit-Track AI
Real-Time Exercise & Calorie Prediction System

Submitted to: Dr. Shikha Purwar Mam

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Problem statement:

- Many people perform exercises at home without tracking their actual performance.
- Uncounted or incorrect repetitions lead to ineffective workouts and poor results.
- Most fitness applications provide only general calorie estimates and ignore exercise-specific data.
- There is a need for a system that can **count exercise repetitions and predict calories burned based on the performed exercise in real-time**.

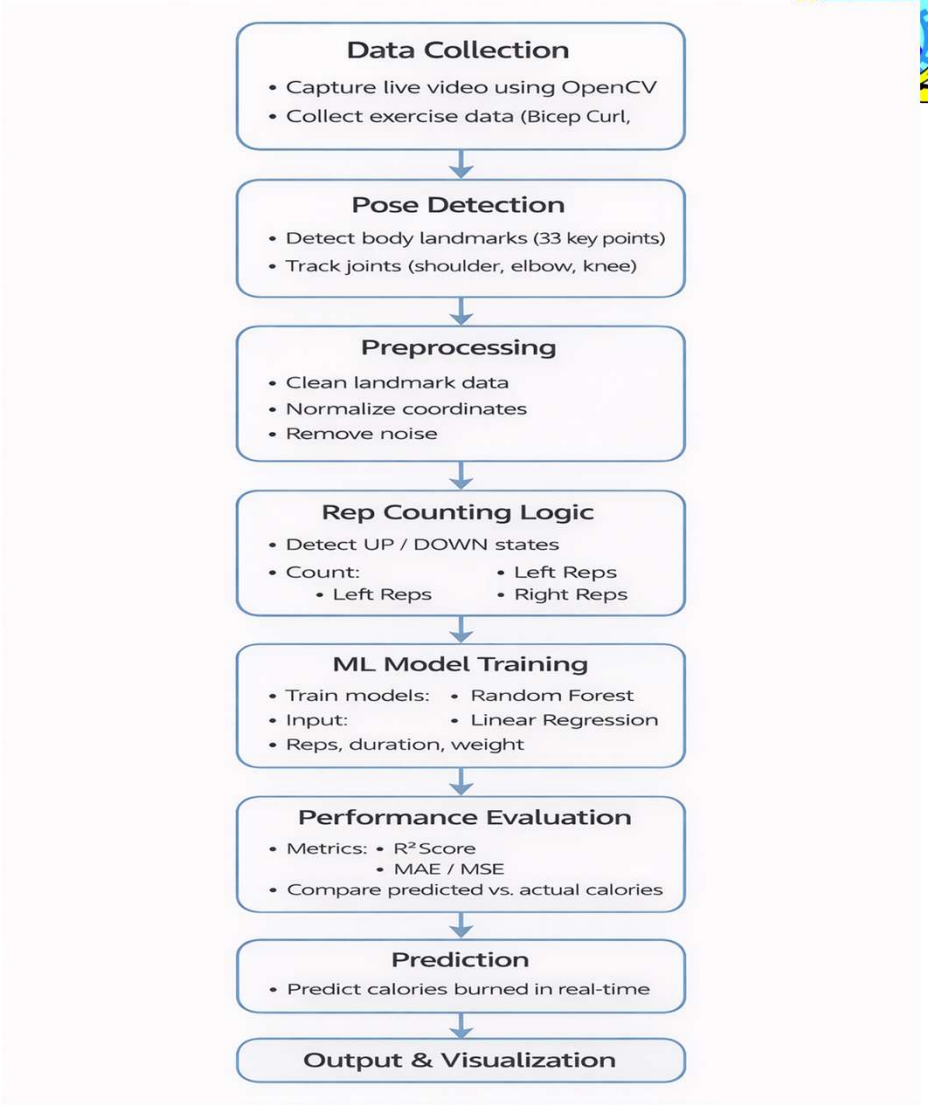


How we are solving the problem?

- We propose a **Fit-Track** using computer vision and machine learning.
- The system uses a **webcam** to capture user exercises in real time.
- It detects body joints using **pose estimation (MediaPipe / BlazePose).**
- Identifies key **body landmarks** such as shoulders, elbows, hips, knees, and ankles.
- Calculates **joint movements** to track exercise performance accurately.
- Automatically **counts exercise repetitions** based on motion patterns.
- **Predicts calories burned** based on the performed exercise and reps.
- Provides **real-time feedback** to improve workout effectiveness.

Project Phases:

Phase 1: Data Collection
Phase 2: Preprocessing
Phase 3: Feature Extraction
(Rep counting Logic)
Phase 4: ML Model Training
Phase 5: Performance Evaluation
Phase 6: Prediction

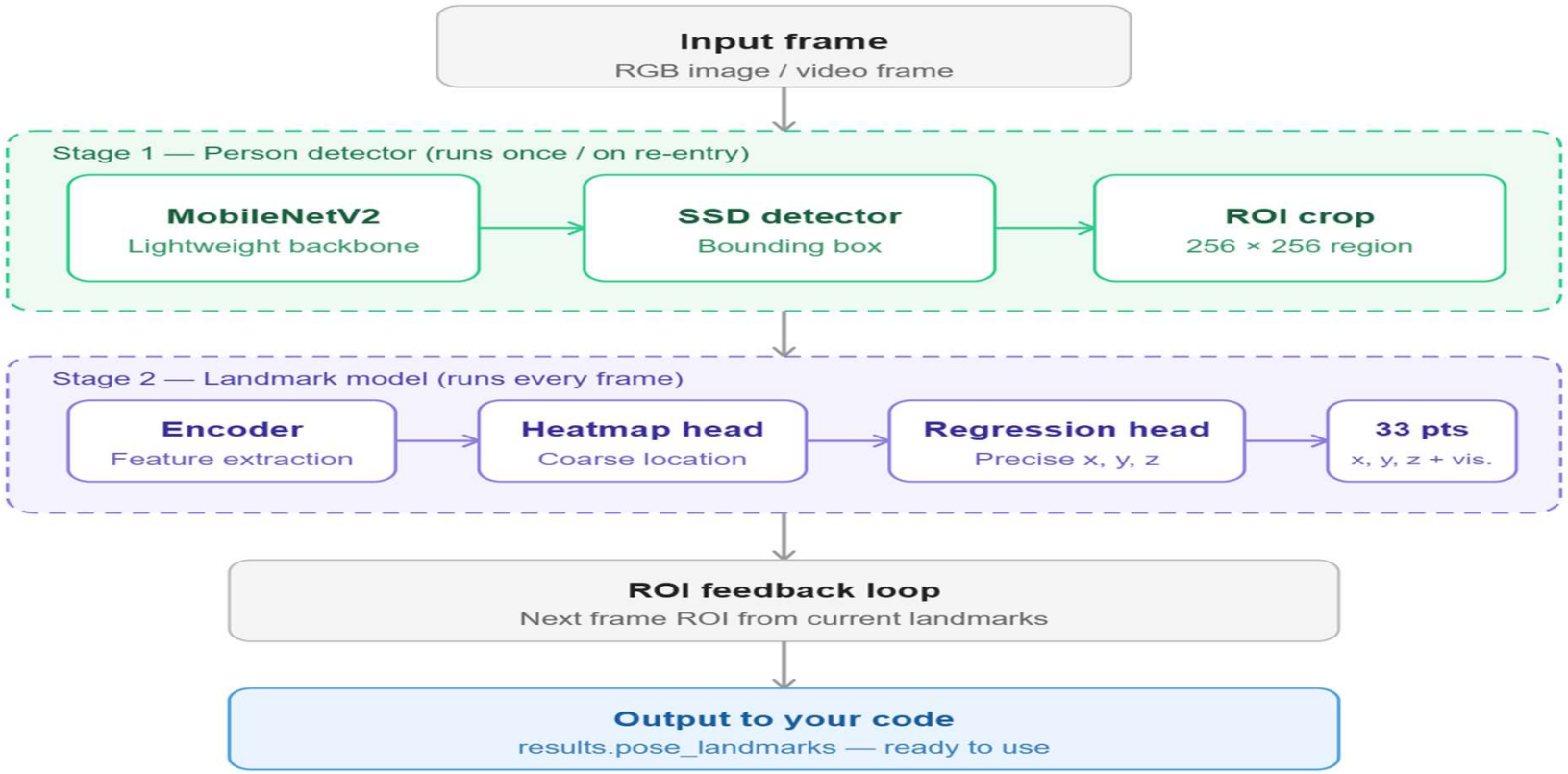


Rep Counting Architecture

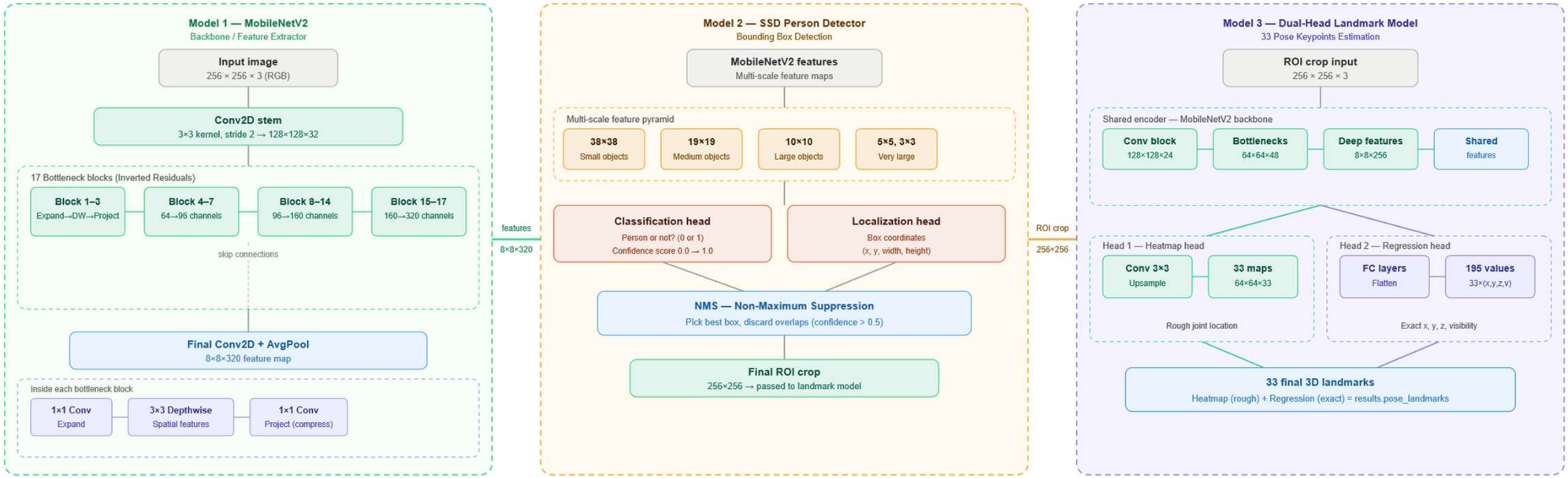


• **Input Processing (OpenCV – Video Capture):**
Captures live video of the user performing exercises through a webcam.
• **Pose Detection (Mediapipe/Blazepose Model):**
Detects 33 body landmarks such as shoulders, elbows, hips, knees, and ankles.
• **Feature Extraction (NumPy):**
Calculates joint angles using the coordinates of detected body landmarks (e.g., elbow angle for bicep curl, knee angle for squat).
• **Rep Counting Logic:**
Analyzes angle changes to detect movement stages (**UP and DOWN positions**).
A repetition is counted when a full movement cycle (DOWN → UP → DOWN) is completed.
• **Output (Rep Counter):**
Displays the number of repetitions performed for each exercise (Left reps, Right reps, Squats).

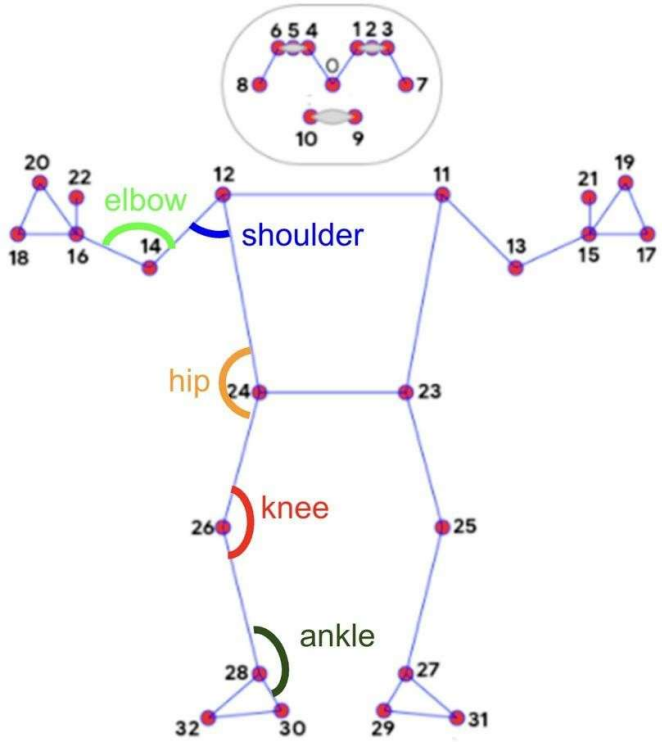
Mediapipe Architecture:



Mediapipe(BlazePose) PipeLining:



Body Landmarks and Joint Angle Calculation:



Rep Counter Mechanism:

- **Pose Detection (MediaPipe):** Detects 33 body landmarks and provides (x, y) coordinates of joints
- **Angle Calculation:**

  $$\theta = |\text{atan2}(y_3 - y_2, x_3 - x_2) - \text{atan2}(y_1 - y_2, x_1 - x_2)| \times 180/\pi$$

- **Rep Counting Logic:**

  If angle > 150° → DOWN (arm straight)

  If angle < 60° → UP (arm bent)

- **Repetition Rule:**

  DOWN → UP → DOWN = 1 rep

- **Output:** Displays total repetitions in real-time

Burned Calories prediction:

- **Input (Exercise Data):** Takes reps, exercise type, and duration from rep counter
- **User Details:** Includes weight, age, height, and gender
- **Feature Preparation:** Combines all inputs into a structured format
- **Models Used:**

1. Random Forest
2. Linear Regression
3. XGBoost
4. SVM

- **Prediction:** Model predicts calories burned based on input data
- **Output:** Displays estimated calories burned

Datasets:

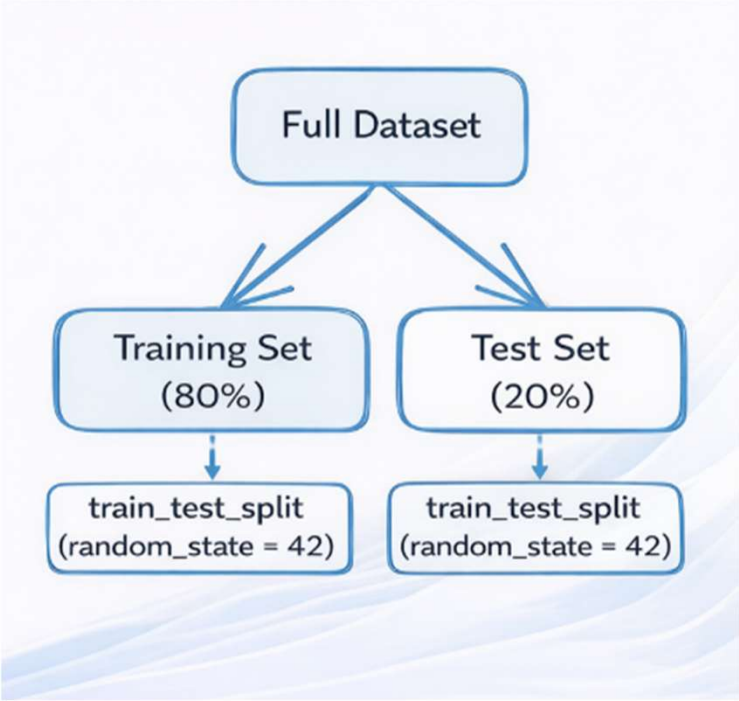
- **Dataset Name:** Exercise & Calories Burn Dataset
- **Total Samples:** 10000 records
- **Number of Features:** 6
  (exercise_type, intensity, body_weight_kg, duration_secs, reps, calories_burned)
- **Input Features:** exercise_type, intensity, body_weight_kg, duration_secs, reps
- **Target Variable:** calories_burned

Dataset Splitting Strategy:

➤ The dataset was split into two mutually exclusive subsets:

Training set: 80%

Test set: 20%

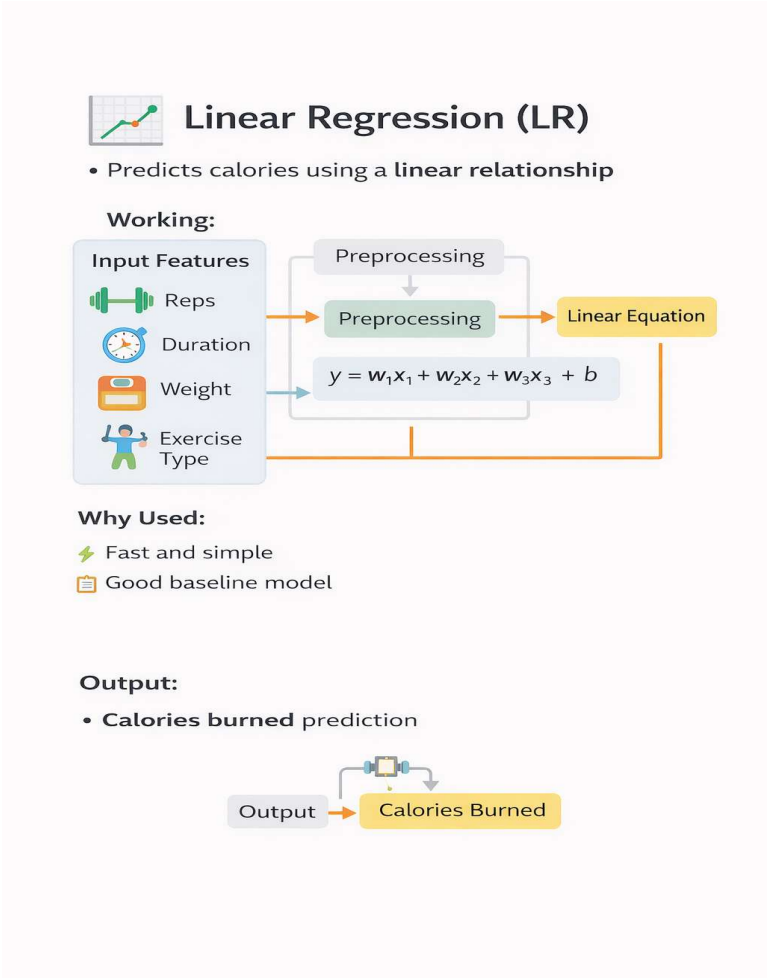


➤ **Key Properties of the Split:**

- Dataset split using **train_test_split**
- Random splitting with **random_state = 42**
- Training set used for **model learning**
- Test set used for **final evaluation**

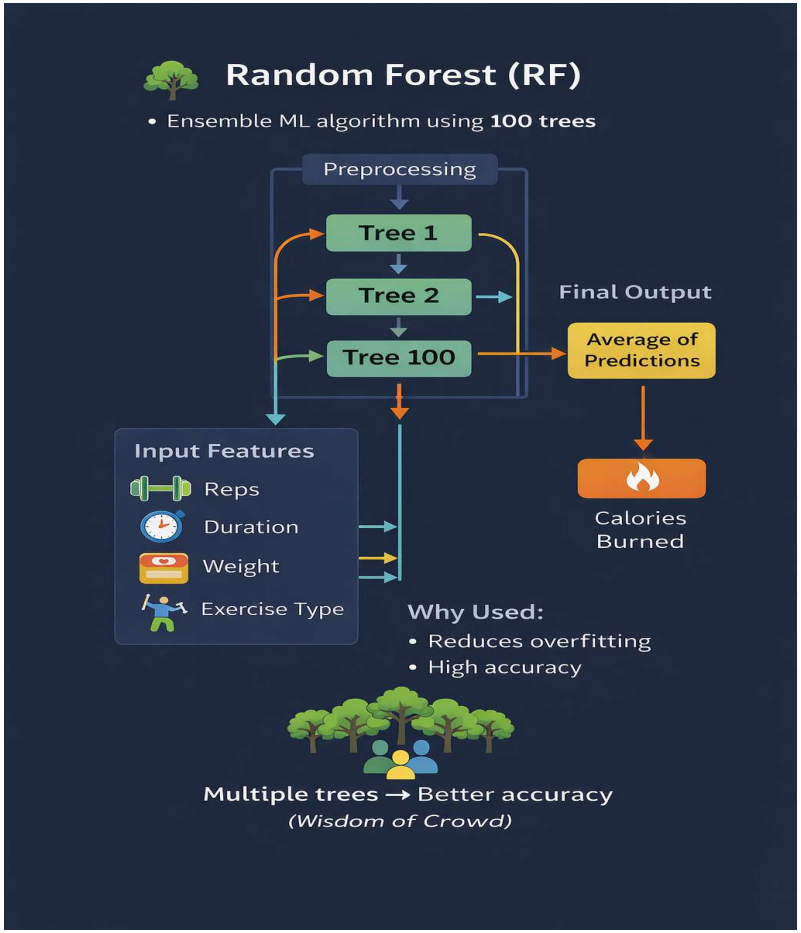
Linear Regression Alogorithm:

- Predicts calories using a **linear relationship**
- Working:
  - Input → Preprocessing → Linear equation
  - Directly maps features to output
- Why used:
  - Fast and simple
  - Good baseline model
- Output:
  - Calories burned prediction



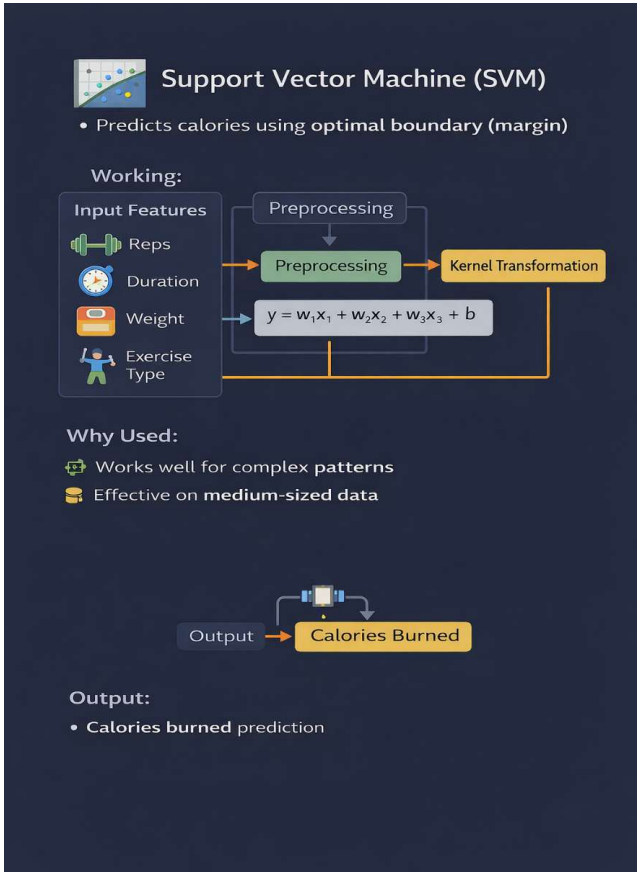
Random Forest Algorithm:

- Ensemble ML algorithm using **100 trees**
- Working:
  - Input → Preprocessing → Multiple trees
  - Each tree predicts calories
  - Final output = average of all predictions
- Why used:
  - Reduces overfitting
  - High accuracy
- Output:
  - Calories burned prediction



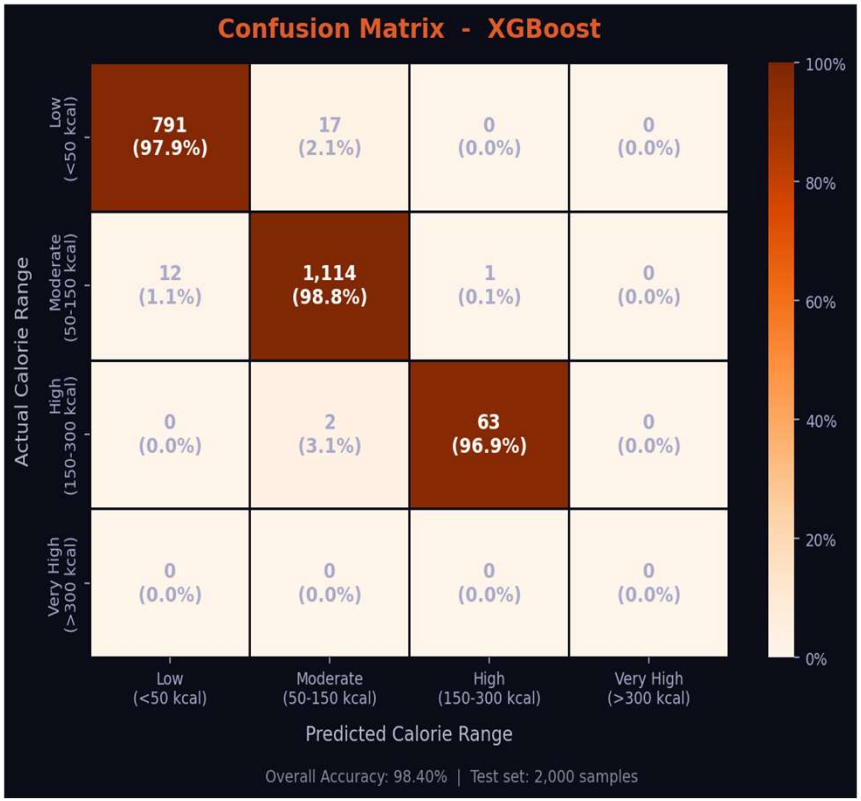
Support Vector Machine (SVM):

- Predicts calories using **optimal boundary**
- Working:
  – Input → Preprocessing → Kernel transformation
  – Finds best fit line/curve for prediction
- Why used:
  – Works well for complex patterns
  – Effective on medium-sized data
- Output:
  – Calories burned prediction

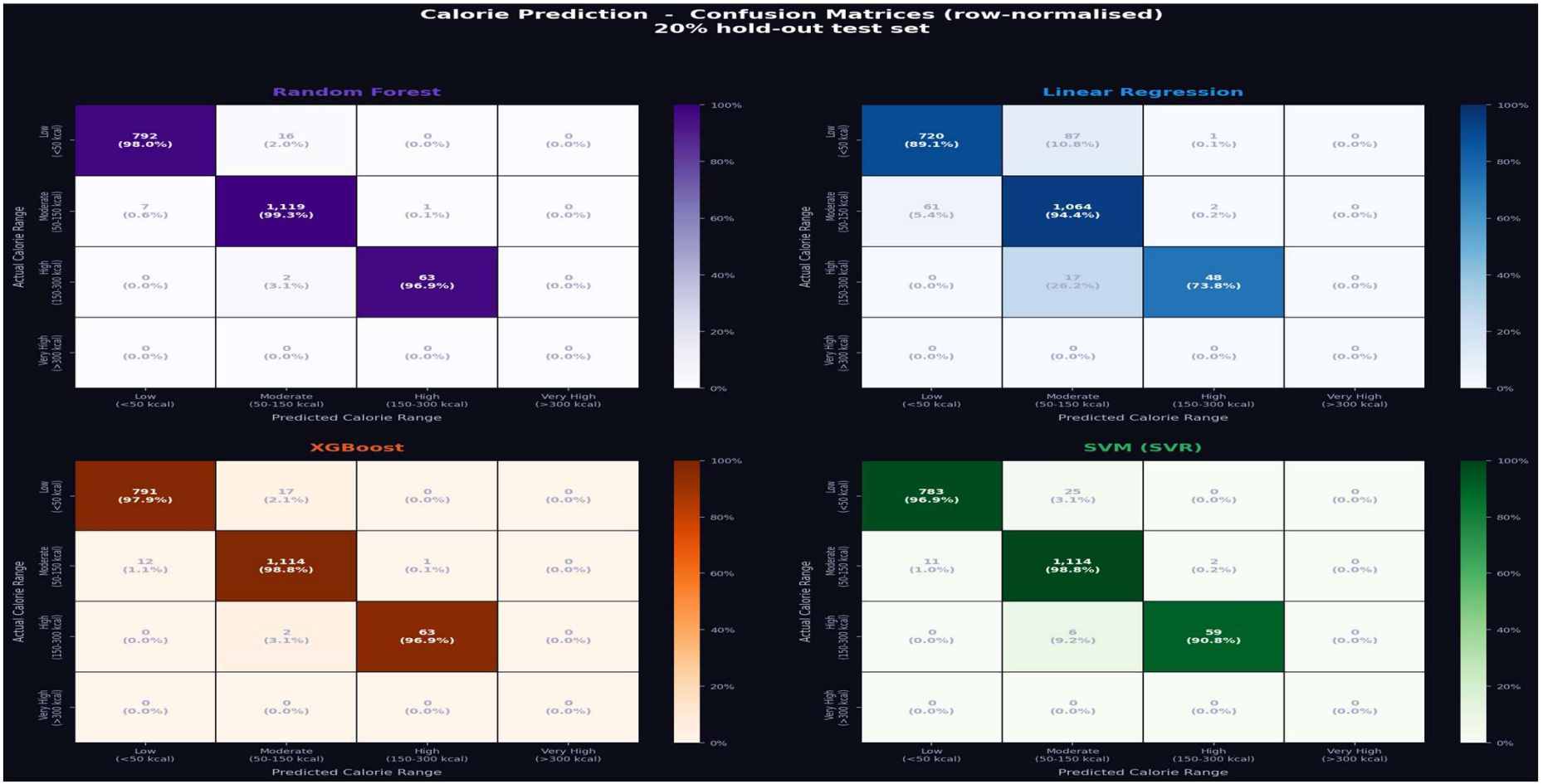


XGBoost (Extreme Gradient Boosting)

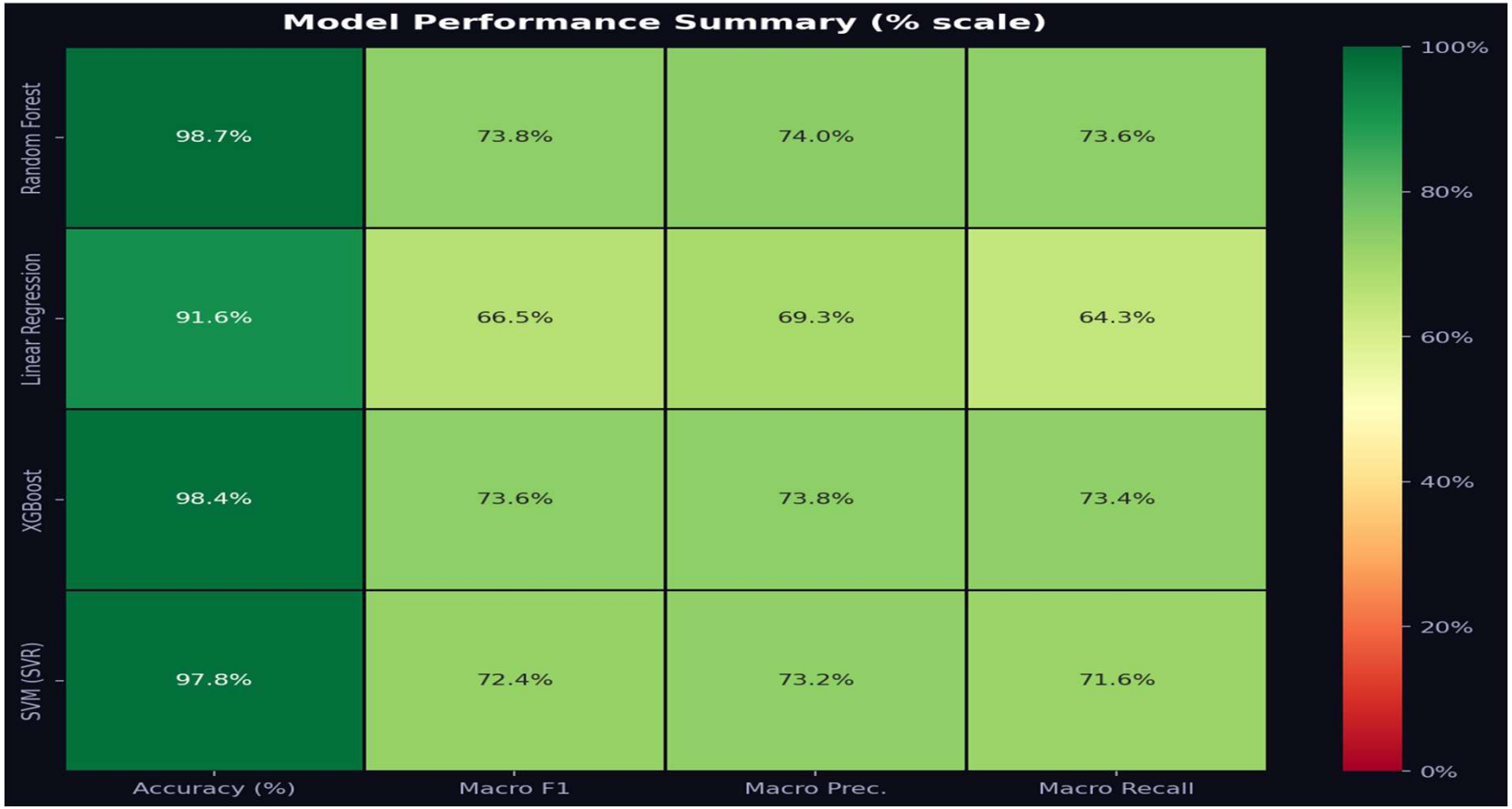
- Predicts calories using **boosted decision trees**
- Working:
  – Input → Preprocessing → Sequential trees
  – Each tree improves previous errors
- Why used:
  – High accuracy
  – Handles complex patterns well
- Output:
  – Calories burned prediction



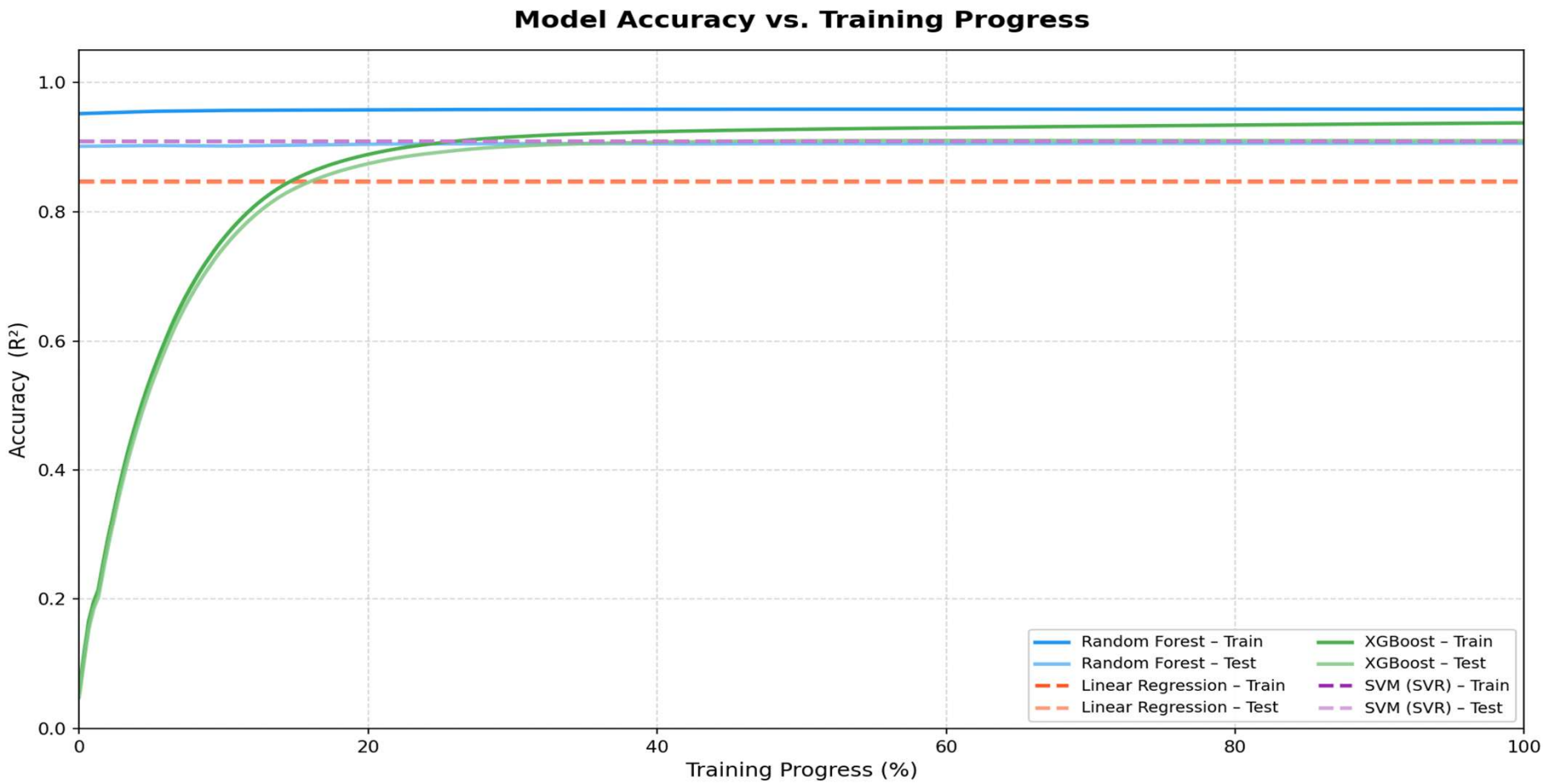
Confusion Matrices:



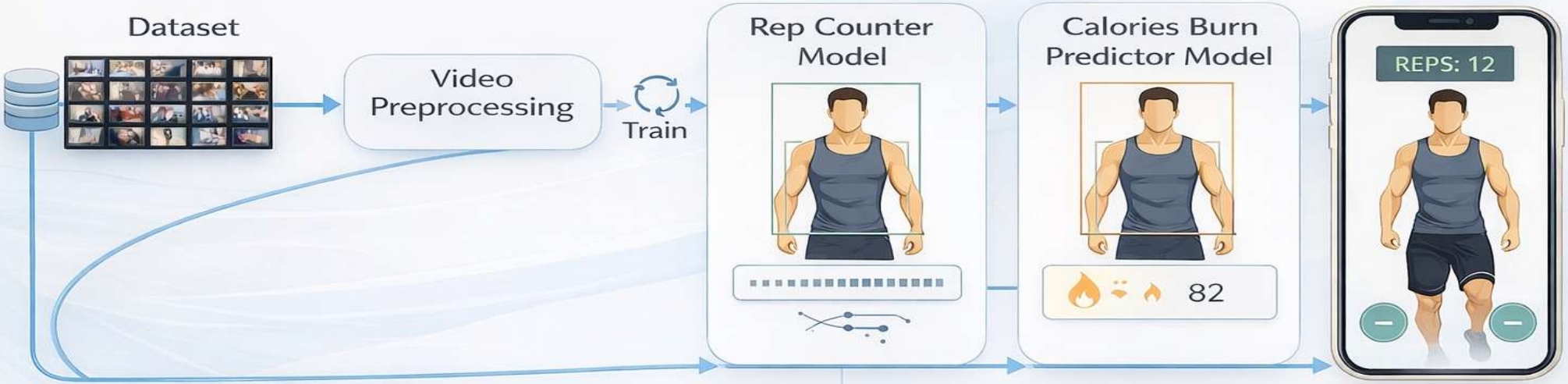
Model Summary Heatmap



Accuracy:


Model Accuracy vs. Training Progress
Accuracy (R²)
Training Progress (%)
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
Random Forest – Train
Random Forest – Test
Linear Regression – Train
Linear Regression – Test
XGBoost – Train
XGBoost – Test
SVM (SVR) – Train
SVM (SVR) – Test

Our project pipeline:



Technology & Techniques Used:

- **Language:** Python
- **Libraries:**
  OpenCV, MediaPipe, NumPy, Pandas
  Scikit-learn (Random Forest, Linear Regression, SVM)
  XGBoost, Matplotlib, Joblib
- **Techniques:**
  Pose Estimation, Angle Calculation
  Real-Time Motion Tracking, Rep Counting
- **Models:**
  Linear Regression, Random Forest, XGBoostSVM

- **Modules:**
  Rep Counter + Calorie Burn Predictor

Future Scope:

- The system can be extended to include **more exercises** such as push-ups, lunges, and jumping jacks for a complete fitness solution
- A **user tracking feature** can be added to store daily workout data like reps and calories burned
- The project can provide **progress analysis over time** to help users understand their improvement
- **Graph-based visualization** can be implemented to display workout history, calories burned, and performance trends
- The system can be enhanced with **personalized recommendations and goal setting** for better user experience

References:

1. Swain, D., Satapathy, S., Acharya, B., Shukla, M., Gerogiannis, V. C., Kanavos, A., & Giakovis, D. (2022).
   Deep Learning Models for Yoga Pose Monitoring. Algorithms, 15(11), 403.Description: This paper proposes a deep learning system for yoga pose recognition using MediaPipe keypoints and a CNN–LSTM model. It detects human poses from video frames and provides feedback for posture correction.
2. MediaPipe Documentation – Google Developers.Description: MediaPipe Pose provides a real-time pose estimation framework that detects 33 body landmarks from video frames. It is used in this project to extract body joint coordinates for exercise tracking.
3. OpenCV Documentation.Description: OpenCV is an open-source computer vision library used for image and video processing. In this project, it is used for camera input, frame processing, and visualization of pose detection.